MIT805 Assignment Part 1: Aeolus MFDD Multi-Modal Flight Delay

Group 4- u23056534 & u20426799

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GitHub Repository: https://github.com/unaizahhhhh/MIT805-Group-4-Assignment.git

1. Introduction

The Aeolus: A Multi-Structural, Multi-Modal, Flight Delay Dataset used in this assignment is publicly available on Kaggle. The data originates from the U.S. Department of Transportation's Bureau of Transportation Statistics. It is published for research and educational purposes and consists of multi-modal aviation data.

The dataset is licensed for open academic use under Kaggle's standard terms of service and is released under the CC BY-NC 4.0 licence (Xu et al., 2025). The licence conditions permits the sharing and adaptation of the dataset provided that appropriate attribution credit is given to the author and not used for commercial purposes. Therefore, this dataset is appropriate for academic analysis in transport, operations research, and distributed data processing.

There is no personally identifiable information included that may violate the POPIA or GDPR and only contains public operational flight data.

This dataset is multi-modal and consists of three major components, tabular CSV data, sequential PyTorch files and graph files. The uncompressed file size is approximately 44GB and the Tabular CSV data is approximately 15GB, however once the data is cleaned it is expected to fall within the size requirements.

2. Dataset Size and Characteristics

The dataset is characterised in the summary *Table 1* below.

Table 1: Summarised Characteristics on the Multi-Structural Flight Delay Dataset

Attribute	Details
Dataset	Aeolus: A Multi-Structural Flight Delay Dataset
Source	I. Flight and airport data: U.S. Department of Transportation, Bureau of Transportation Statistics (BTS, n.d.). II. Weather data: Meteostat, n.d.
Size – Raw Dataset	44.08 GB (Uncompressed)
Size – Working Dataset	14.49 GB (Flight_Tab)
Size – Processing Dataset	14.49 GB
Format	Multi-structural; CSV tabular
Data Collection Period	9 Year period; 2016 - 2024
Publication Date	18 September 2025
Last Update Date	23 April 2026

3. Data Quality and Preparation

I. The dataset was checked for missing values. The flight operation variables contained no missing values. However the weather variables experienced some missing values which accounted for <0.03%. The dataset also contained no specific variable that identified cancelled flights.

II. The data was checked for duplicate entries; however, no duplicates were present.

III.

IV. Descriptive statistics were used to evaluate the numerical variables for potential outliers. Some observations contained excessively high values, particularly for departure delay, taxi times, and weather data. These require further investigation to determine whether they are extreme but legitimate values or simply data-quality errors.

V. The variables were evaluated for non-commercial flight operations such as test flights, however these were not indicated in the information.

VI. The data is related to US commercial air travel and is not a representative of the global scenario. Furthermore, an airline that contributes to a higher delay number may have more operations. Thus, the percentage contribution to its own fleet will be a reasonable measure to assess.

VII. The data may contain temporal bias and distribution drift such as seasons, policy changes, operational changes (Xu et al., 2025) and holiday periods, and weather phenomena.

VIII. The dataset contains a few categorical entries such as destination, origin, and carrier. These need to be encoded to numerical values when processing the data.

IX. Data leakage is a concern, as information of the outcome could have been included before its outcome was known.

X. The dataset used does not include any cancellation variable. Furthermore, cancellation status cannot be determined from the data presented.

4. Exploratory Data Analysis

I. The dataset is composed of 54.7 million flights, with the potential of providing a comprehensive analysis of the U.S. domestic air traffic performance. According to the Bureau of Transportation Statistics (BTS, n.d.) a flight is classified as a delay if it runs 15 minutes behind the scheduled departure or arrival time. The latter was used in the analysis. Across all operational data, 18.5% of flights experienced delays of 15 minutes or more, while 81.5% of flights arrived on time or earlier, as per *Table 2* in the Appendix. Although this suggests that the majority of flights operate reliably, there still remains a considerable number that experience delays. The distribution of delays is skewed towards the right as can be seen in *Figure* 1 in the Appendix. As the delay times increase the number of affected flights decreases significantly towards zero, with a small number showing excessive delays. Most flights however arrive slightly before schedule.

II. Seasonal changes have a strong influence on flight delays. Refer to *Figure 2* in the Appendix. The lowest average delays occur in September, October and November, this indicates stable autumn operations and less travel congestion. Delay times increase sharply during June and July. This is in alignment with the seasonal challenges of summer rainfall, higher passenger volumes due to summer vacation, and congestion. Winter months display mixed behaviour with increased delay times in December and January that is most likely due to the holiday season and winter weather disruptions such as snowfall. These patterns demonstrate that flight performance is strongly associated with seasonal changes.

III. It is also evident that there were a significant reduction in flight volumes during the time of the Covid-19 pandemic in 2020 as is illustrated in *Figure 3 (see Appendix)*. This relieved congestion and resulted in flights arriving 5 minutes earlier for the year at the destination.

IV. Furthermore, rainfall impacts flight punctuality as per *Table 3*, with flights experiencing precipitation at the departure point or destination showing higher mean delay times. Flights in dry conditions have only an average delay of 3.7 minutes versus a 15-minute delay under precipitation.

V. Airline performance also varies across carriers as illustrated in *Figure 4 (in the Appendix)*. Airlines such as Jet Blue (B6) and Frontier (F9) exhibit the highest delay frequencies, exceeding 25%. An interesting observation is that although some airlines have a smaller footprint, they contributed to higher delay rates than most bigger carriers. Examples include Virgin America (VX) and Allegiant Air (G4) who combined constituted only 0.84% of the flights yet had delayed frequencies of 23% each. In contrast, legacy airlines such as Delta (DL), United (UA), and American (AA) show moderate delay levels ranging between 14.5% and 19.2%. Overall the ranking highlights meaningful insights around operational reliability, with some airlines outperforming others.

VI. The routes associated with the most delays reveal extreme operational anomalies as per *Figure 5* in the Appendix. This was observed in the distribution plot, *Figure 1*. Several airport pairs show average delays exceeding 700–1200 minutes, far beyond typical operational ranges. Routes such as Akron-Canton OH (CAK) to McGhee Tyson Airport TN (TYS), and Hollywood Burbank (BUR), Los Angeles, CA to Fort Lauderdale-Hollywood, FL (FLL) stand out as extreme cases with delays of 1237 and 922 min recorded respectively. Most of these results are based on one or two delayed flights thus the results must be viewed strictly as abnormalities. These findings emphasize the importance of identifying and investigating route-specific anomalies when analyzing large-scale flight datasets.

VII. On the positive side, certain airlines also outperformed others with higher on-time arrivals recorded. Refer to *Table 4 in the Appendix*.

5. The 7 V's of the Dataset

I. **Volume:** The Aeolus dataset consists of 54.7 million static historical flight records. Over multiple domestic flight operations high volumes of flight and weather data were generated on a continuous basis and were consolidated in batches. which requires high computational capacity. Thus the importance of using PySpark is emphasised.

II. **Velocity:** Although the dataset is static and historic, velocity relates to the computational speed at which data processing is required. In a deployment scenario, flight and weather data can be streamed to predict potential delays and feed the information to a passenger whilst making an online flight booking, flag the delay at the ATC tower, and inform operations to make changes on passengers secondary flights.

III. **Variety:** The dataset is comprised of various information categories such as geographic/spatial/route data (departure and destination locations), flight operational data (departure and arrival dates and times, taxi in and out times), weather data (departure- and destination airport temperatures, precipitation, wind speeds), temporal data (week, month), and airline carrier or flight number. These factors are important variables to be considered in order to produce reliable predictions.

IV. **Value:** The value can be found in the application of the learnings. Passengers can make informed decisions about post-trip arrangements, air traffic controllers can make provisions, and airport operations can make appropriate adjustments to connecting flights and plan resources. This drives airport economics as airlines can, according to Wu et al., better optimise operations and fuel consumptions related to longer taxiing, and compensation for delayed passengers. The social impact can be realised in the improved satisfaction of the traveller and timely rebooking of other flights.

V. **Veracity:** A very small percentage of weather related data are missing from the dataset and this incompleteness introduces as small uncertainty. Despite this over 99.97% of data are complete and viewed as reliable.

VI. **Variability:** It has been demonstrated that the data changes over time periods. Seasons, weather conditions and route dynamics can influence results.

VII. **Visualisation:** The dataset has high potential for visualised display on graphs for dashboards. This can be shown with flight delays over days of the week, months of the year, by airline or carrier, or a comparative diagram of wet vs dry conditions. Refer to Figure 1 to Figure 5 in the Appendix. Furthermore flight patterns can be mapped geographically for different routes. Aggregated results from MapReduce can be linked to the spatial map to identify regional patterns for delays as well as bottlenecks.

6. Business and Societal Value and Limitations

The operation flight data provides strong business value for both airlines and airports. By analysing delay patterns across airlines, routes, locations and time periods, this can aid airlines in identifying inefficiencies and drive optimisation efforts in resource management such as scheduling, aircraft usage and staff rotations. Furthermore, it may also help airlines in cost reduction and drive airline reliability. Insights such as on–time performance rankings allow airlines to benchmark themselves against their competitors and make targeted improvements to enhance their business performance. Insights can be given to passengers while making online bookings or to airport operations to allow for sufficient rescheduling of connecting flights. Airports can optimise operational costs by reducing compensation for delayed passengers. The societal value is found in the unlocked satisfaction of passenger experience.

The dataset also provides information that may be of societal benefit. Understanding how weather conditions, seasonality, and travel congestion affect delays can help improve user experience by reducing wait times and missed connecting flights. Government and private entities can use these insights to direct where future improvements and investments can be made in terms of infrastructure. Additionally, reducing delays can contribute to lowering fuel consumption and emissions, supporting environmental sustainability.

The dataset contains a few limitations which may affect the reliability of certain insights. Weather information containing precipitation levels is often missing or inconsistently recorded across different airports, which reduces the accuracy of analysis related to weather conditions. In addition, the presence of extreme outliers suggests irregular operations such as airport closures, severe weather events, or misclassified cancellations on small occasions. These anomalies can distort averages or predictions and make some route-level or seasonal results appear more severe than typical operational behavior and needs to be considered.

7. References

Bureau of Transportation Statistics. (n.d.). Reporting carrier on-time performance (1987–present). U.S. Department of Transportation. Retrieved September 2, 2026, from https://www.transtats.bts.gov/TableInfo.asp? QO_fu146_anzr=b0-gvzr&gnoyr_VQ=FGJ

Meteostat. (n.d.). *Meteostat developers: Historical weather and climate data*. Retrieved September 2, 2026, from https://dev. meteostat.net/

Xu, L., Yuan, X., Liang, Y., Yin, S., & Wu, Y. (2025). Aeolus: A multi-structural flight delay dataset. Advances in Neural Information Processing Systems, 38. https://doi.org/10.52202/085713-2048

8. Appendix

Table 2: Overview of Flight Delays in the Dataset

Flight Status	No. of Flights	Percentage (%)
Delayed (≥15 min)	10,116,916	18.50%
Not Delayed (<15 min)	44,557,087	81.50%
Total	54,674,003	100%

Table 3: Average Delays with and without Precipitation

Precipitation	Average Arrival Delay (min)	
	With Precipitation at Origin	With Precipitation at Destination
No	3.68	3.66
Yes	15.12	15.23
Missing Info	4.3	4.21

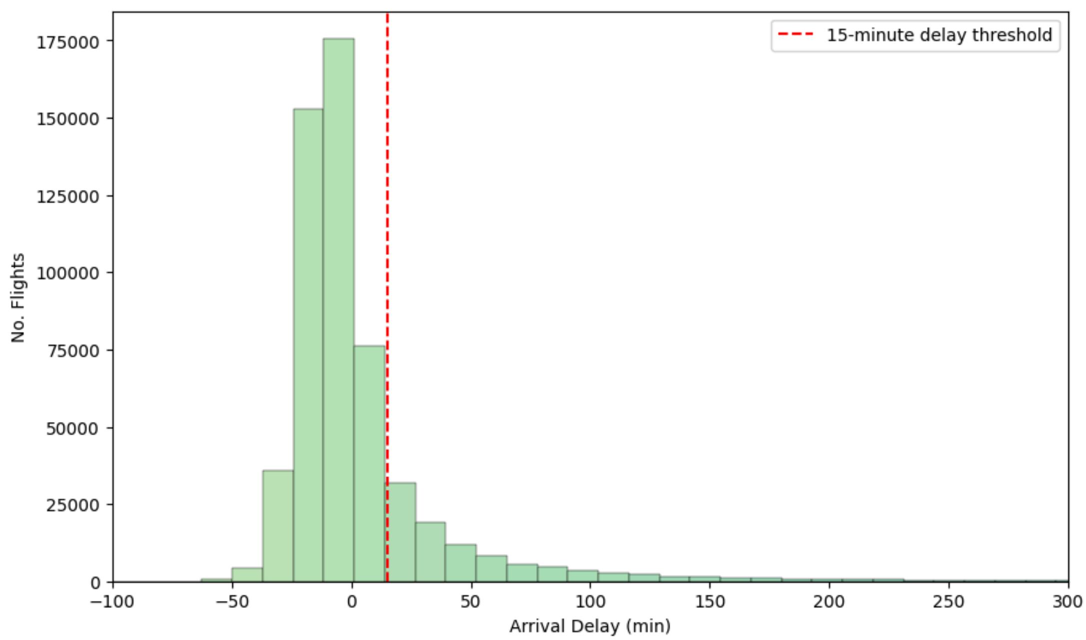


Figure 1: Distribution of Flight Delays on Small Sample from Dataset

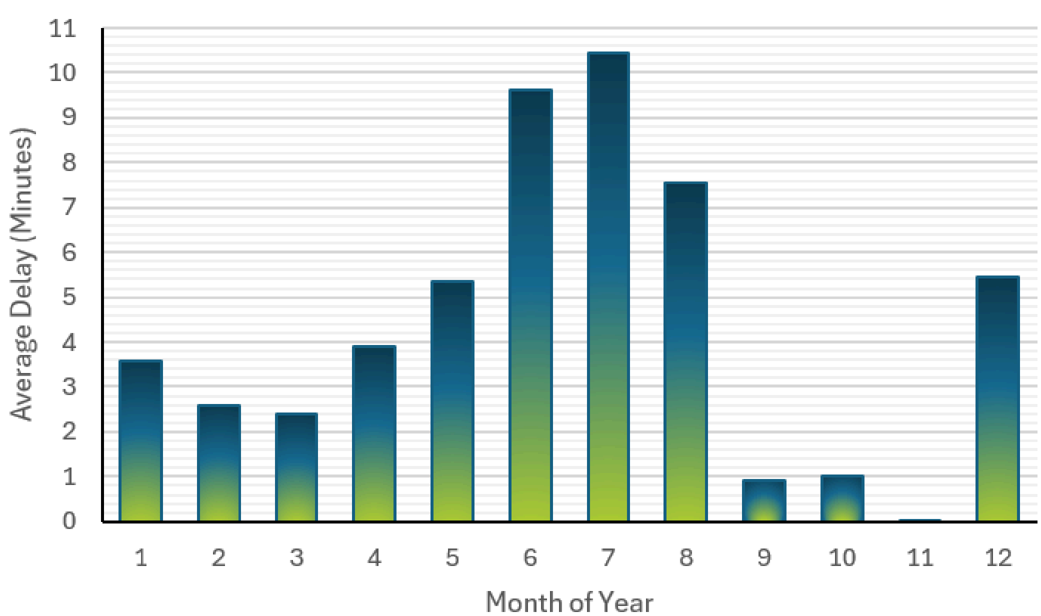


Figure 2: Average Delay per Month

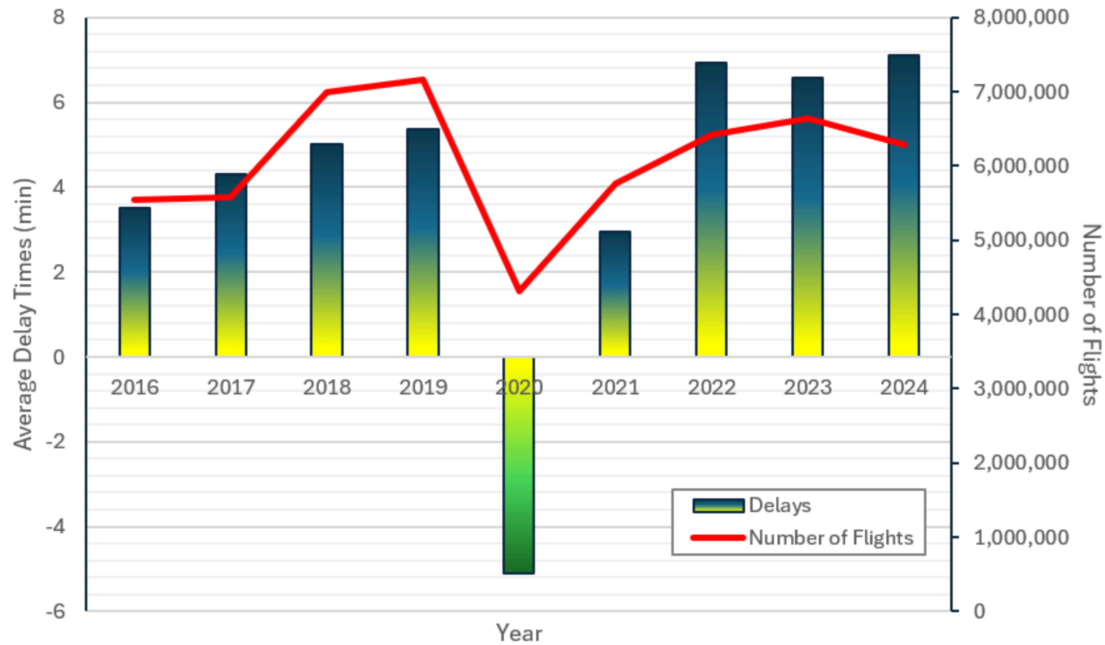
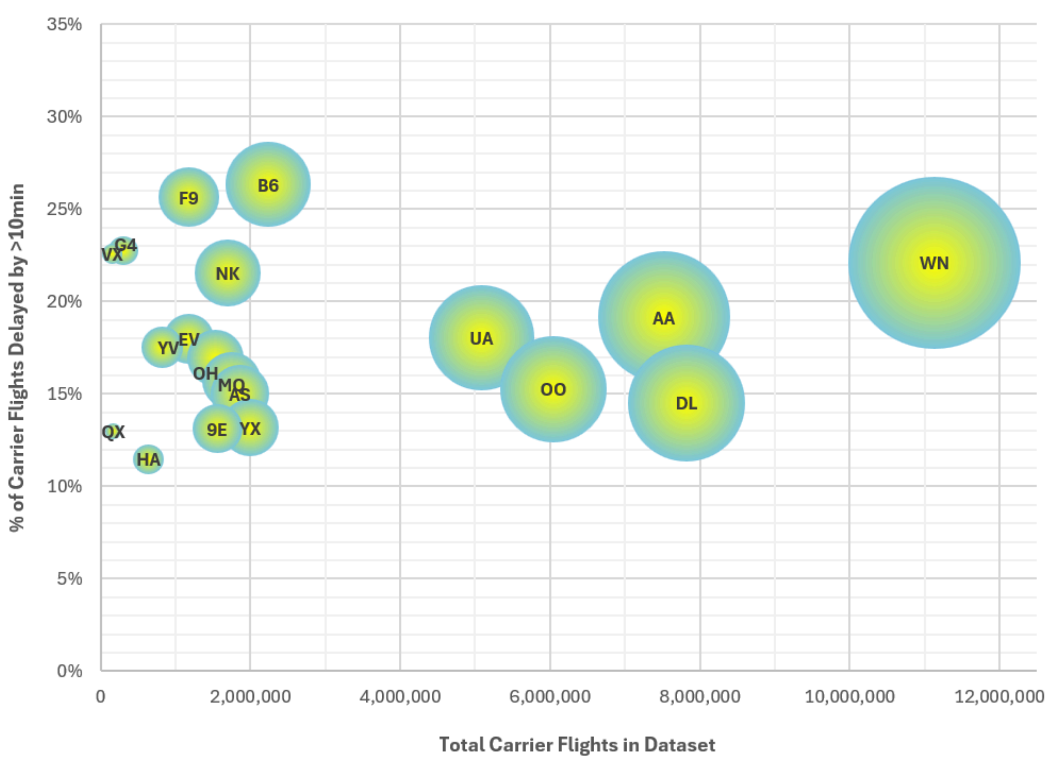


Figure 3: Average Delay per Year



B6	JetBlue Airways
F9	Frontier Airlines
G4	Allegiant Air
VX	Virgin America
WN	Southwest Airlines
NK	Spirit Airlines
AA	American Airlines
UA	United Airlines
EV	ExpressJet Airlines
YV	Mesa Airlines
OH	PSA Airlines
MQ	Envoy Air
OO	SkyWest Airlines
AS	Alaska Airlines
DL	Delta Air Lines
YX	Republic Airways
9E	Endeavor Air
QX	Horizon Air
B6	JetBlue Airways

Figure 4: Flight Delays by Carrier Plot

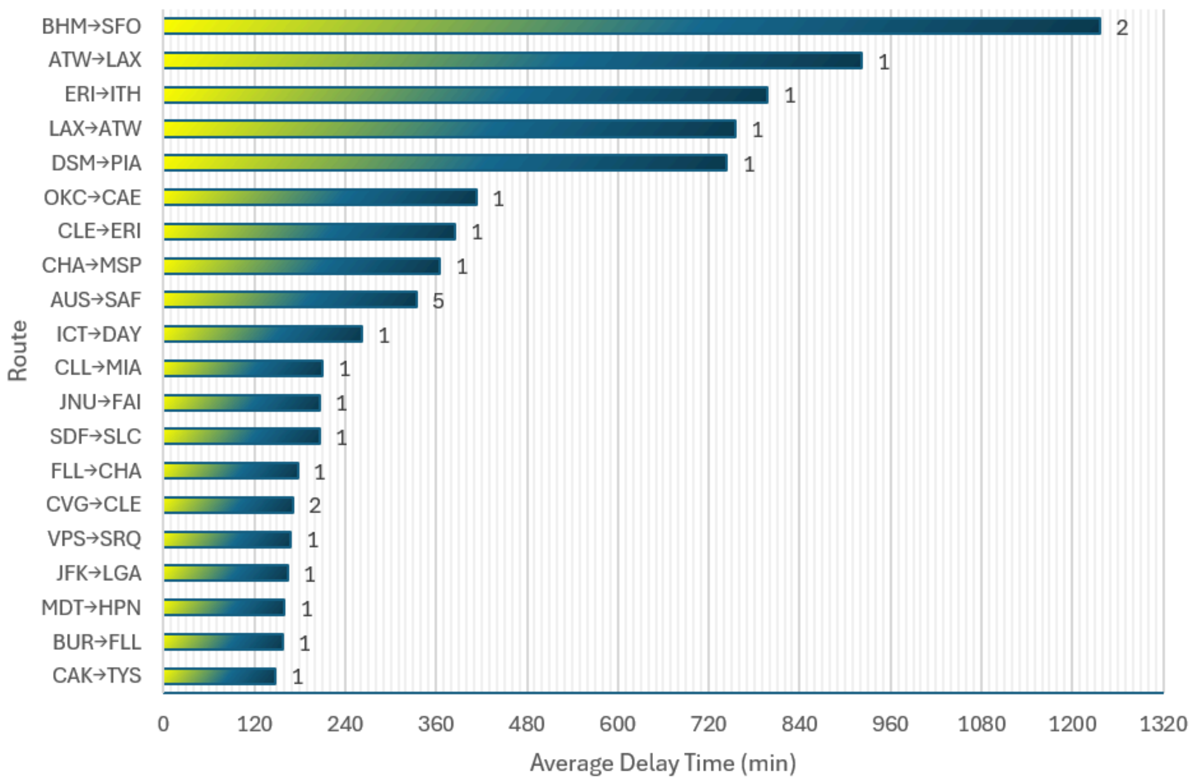


Figure 5: Top 20 Delayed Routes (Number of flights indicated in the data labels)

Table 4: On-time Performance Ranking by Airline Carrier

Carrier	On-Time Rate (%)	Number of Flights
9E	75.44%	1,558,088
DL	71.92%	7,821,658
YX	71.44%	1,995,607
OO	68.29%	6,044,308
UA	67.08%	5,083,571
EV	66.22%	1,178,841
OH	65.58%	1,538,083
YV	65.40%	818,626
MQ	65.37%	1,747,132
QX	65.19%	169,282
AS	63.70%	1,859,574
AA	63.38%	7,518,473
NK	62.85%	1,696,082
WN	62.55%	11,133,643
HA	60.90%	638,962
B6	59.38%	2,236,608
F9	59.01%	1,176,549
G4	57.74%	303,824
VX	55.12%	155,092

Table 6: Top 20 Delayed Routes

Origin	Destination	Average Arrival Delay (Minutes)	No. of Delays
CAK	TYS	1237	1
BUR	FLL	922	1
MDT	HPN	798	1
JFK	LGA	755	1
VPS	SRQ	744	1
CVG	CLE	413	2
FLL	CHA	385	1
SDF	SLC	365	1
JNU	FAI	334	1
CLL	MIA	262	1
ICT	DAY	210	1
AUS	SAF	206	5
CHA	MSP	206	1
CLE	ERI	177	1
OKC	CAE	171	1
DSM	PIA	168	1
LAX	ATW	164	1
ERI	ITH	160	1
ATW	LAX	157	1
BHM	SFO	148	2

Table 5: Flight Delays by Carrier

Carrier	Total Flights	Delayed Flights	Delayed Percentage (%)
B6	2,236,608	588,850	26.3%
F9	1,176,549	301,744	25.7%
G4	303,824	69,106	22.8%
VX	155,092	34,993	22.6%
WN	11,133,643	2,460,739	22.1%
NK	1,696,082	365,310	21.5%
AA	7,518,473	1,440,670	19.2%
UA	5,083,571	918,186	18.1%
EV	1,178,841	211,839	18.0%
YV	818,626	143,496	17.5%
OH	1,538,083	260,583	16.9%
MQ	1,747,132	274,790	15.7%
OO	6,044,308	922,459	15.3%
AS	1,859,574	279,310	15.0%
DL	7,821,658	1,134,684	14.5%
YX	1,995,607	263,116	13.2%
9E	1,558,088	204,493	13.1%
QX	169,282	21,973	13.0%
HA	638,962	73,275	11.5%